Chap 7 Applications of Integration

§2 Volumes

* A : the area of base of the cylinder

h : the heigh of the cylinder

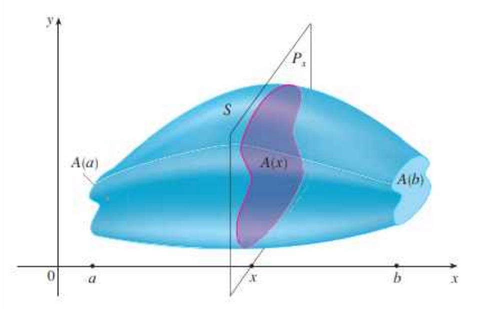
$\Rightarrow$ The volume V of the cylinder is

$$V = Ah.$$

* <u>Definition of Volume</u>

S : a solid that lies between $x=a$ and $x=b$

$A(x)$: the area of the cross-section of S in a plane P_x, perpendicular to the x-axis and passing through the point x, where $a \le x \le b$.



$\Rightarrow$ The volume V of S is

$$V = \lim_{n\to\infty}\sum_{i=1}^{n} A(x_i^*)\,\triangle x.$$
$$= \int_a^b A(x)\,dx$$

<u>Exam 1</u> Show that the volume of a sphere of radius r is

$$V=\frac{4}{3}\pi r^3 .$$

(sol)

Exam 2 Find the volume of the solid obtained by rotating about the x -axis the region under the curve $y=\sqrt{x}$ from 0 to 1.
(sol)

Exam 3 Find the volume of the solid obtained by rotating the region bounded by $y=x^3$, $y=8$, and $x=0$ about the y-axis.
(sol)

Exam 4 The region $\Re$ enclosed by the curves $y=x$ and $y=x^2$ is rotated about the x-axis. Find the volume of the resulting solid.
(sol)

Exam 5 Find the volume of the solid obtained by rotating the region in Example 4 about the line $y=2$.
(sol)

Exam 6 Find the volume of the solid obtained by rotating the region in Example 4 about the line $x = -1$.
(sol)

<u>Exam 7</u> Figure 12 shows a solid with a circular base of radius r. Parallel cross-sections perpendicular to the base are equilateral triangles. Find the volume of the solid

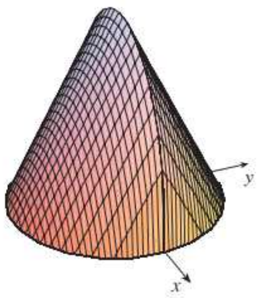


FIGURE 12

(sol)

<u>**Exam 8**</u> Find the volume of a pyramid whose base is a square with side L and whose heigh is h.
(sol)